6-4-1: Examining Wireless Communications

At the end of this episode, I will be able to:

1. Identify and explain wireless communication technologies, given a scenario.

Learner Objective: *Identify and explain wireless communication technologies, given a scenario.*

Description: In this episode, the learner will learn various wireless communication technologies and standards. We will explore cellular, satellite, Bluetooth, and infrared.

- What are wireless communication?
    - An unbounded form of transmission media
    - Uses radiated energy to transmit signals
    - Examples:
        - 802.11 Wi-Fi
        - Cellular (CDMA, GSM (2G), UTMS (3G), HSPA/HSPA+ (3/3.5G), LTE (4G), 5G)
        - Satellite
        - Bluetooth
        - Infrared
- What are some cellular communications examples?
    - CDMAone (Code Division Multiple Access)
        - 2G technology
        - Uses code-based multiplexing for multiple users.
        - Allows simultaneous transmission of several communications over a single channel.
    - GSM (Global System for Mobile Communications)
        - 2G technology
        - Utilizes time division multiplexing (TDMA).
        - Widely adopted globally, known for SIM card use.
    - UMTS (Universal Mobile Telecommunications System)
        - 3G standard, offering higher data rates.
    - HSPA/HSPA+ (High-Speed Packet Access)
        - Enhanced 3G/3.5 technology, improving speed and capacity.
    - CDMA2000
        - CDMA-based 3G technology
        - Providing better speeds for multimedia and mobile Internet
        - No SIM, uses Electronic Serial Number (ESN)
    - LTE (Long-Term Evolution)
        - Standard for 4G wireless broadband technology.
        - Offers increased network capacity and speed for mobile and data communications.
        - Uses a different radio interface and core network improvements compared to 3G.
    - 5G Technology:
        - The latest generation of cellular network technology.
        - Provides higher speeds, lower latency, and greater capacity than 4G LTE.
        - Supports a vast network of IoT devices and new applications like autonomous vehicles.
- What is a SIM Card?
    - Subscriber Identity Module card - a small, removable smart card used in mobile devices to store data for GSM, UMTS, and certain LTE networks
- What are some Bluetooth Standards?
    - Operates in the 2.4 GHz ISM band
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- Bluetooth 1.x
    - Basic rate of 1 Mbps.
    - Limited range (10 meters) and data rate
  - Bluetooth 2.x
    - Range around 30 meters
    - Introduction of Enhanced Data Rate (EDR) for faster data transfer up to 3 Mbps.
    - Better resistance to interference.
    - 2.1 version introduced Secure Simple Pairing (SSP) for improved pairing experience.
  - Bluetooth 3.x + HS (High Speed)
    - Range: around 30 meters
    - Introduction of high-speed data transfer, utilizing Wi-Fi for data rates up to 24 Mbps.
    - Continued improvements in power consumption and security.
  - Bluetooth 4.x (Bluetooth Low Energy or BLE)
    - Range: 60 meters
    - Up to 24 Mbps
    - Focus on significantly lower power consumption.
    - BLE technology allowed for long-term operation with battery-powered devices like fitness trackers.
    - 4.2 improved privacy and increased data capacity.
  - Bluetooth 5.x
    - Increased range (around 200 meters), speed (up to 2 Mbps for LE, 50 Mbps for EDR), and broadcasting message capacity.
    - Enhanced focus on IoT (Internet of Things) with improvements in connectivity and range.
    - Multiple simultaneous connections
  - Bluetooth 5.2
    - Improved efficiency and support for LE Audio, including features like Audio Sharing and Broadcast Audio.
    - Enhanced reliability and performance for audio transmissions.
  - Bluetooth 5.3
    - Perioc Advertising Enhancement
    - Encryption Key Size Control
  - Bluetooth 5.4
    - Periodic Advertising with Responses (PAwR)
    - Encrypted Advertising Data
- What are some of the IrDA infrared standards?
  - Line-of-site communication, (3ft/1m) easily obstructed
  - SIR (Serial Infrared): Standard speed up to 115.2 kbps.
  - MIR (Medium Infrared): Intermediate speed, ranging from 0.576 to 1.152 Mbps.
  - FIR (Fast Infrared): Higher speeds, up to 4 Mbps.
  - VFIR (Very Fast Infrared): Very high speeds, up to 16 Mbps.

- Additional Resources:
  - If applicable